

ORANGE-CARITAT AB, France

WMO: 075790

Lat: 44.144N

Lon: 4.861E

Elev: 60

StdP: 100.61

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.2	-1.7	-9.3	1.7	0.3	-7.2	2.1	1.7	17.9	5.6	16.5	4.6	3.9	350	0.534

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.8	34.6	21.0	33.0	20.7	31.5	20.3	22.7	30.9	22.0	30.0	21.2	29.0	3.9	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.3	15.1	25.3	19.4	14.3	24.9	18.6	13.6	24.4	67.0	30.8	64.3	30.2	61.6	29.1	25.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
15.2	13.6	12.2	DB	-5.7	37.2	2.1	1.9	-7.1	38.6	-8.3	39.7	-9.5	40.8	-11.0	42.2	(4)
			WB	-6.3	24.3	2.2	0.8	-7.9	24.8	-9.2	25.3	-10.4	25.7	-12.0	26.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	14.7	6.0	7.0	10.6	13.5	17.6	21.9	24.4	23.9	19.4	15.4	9.8	6.6	(6)
(7)		DBStd	7.21	3.29	3.38	3.10	3.08	3.02	3.27	2.78	2.82	2.97	3.19	3.50	3.41	(7)
(8)		HDD10.0	418	128	91	30	6	0	0	0	0	3	44	116		(8)
(9)		HDD18.3	1886	382	316	240	149	49	7	1	1	22	99	256	364	(9)
(10)		CDD10.0	2139	6	8	48	109	234	356	447	430	282	171	39	9	(10)
(11)		CDD18.3	566	0	0	0	2	26	113	190	173	54	8	0	0	(11)
(12)		CDH23.3	5638	0	0	2	36	263	1150	1988	1701	457	41	0	0	(12)
(13)		CDH26.7	2311	0	0	0	3	56	455	910	758	126	3	0	0	(13)
(14)	Wind	WSAvg	4.6	5.0	5.4	5.0	4.7	4.5	4.5	4.3	4.3	4.1	4.8	4.6	(14)	
(15)	Precipitation	PrecAvg	743	58	40	40	65	59	42	39	42	111	99	48	(15)	
(16)		PrecMax	1203	166	122	102	167	131	120	129	131	343	174	273	212	(16)
(17)		PrecMin	487	0	0	0	13	6	3	1	2	20	24	22	0	(17)
(18)		PrecStd	176	44	33	31	39	35	30	33	28	73	42	65	47	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	16.5	18.5	23.1	26.6	30.9	35.5	36.4	37.3	32.5	26.4	20.1	16.7	(19)
MCWB			11.8	11.5	13.1	16.4	18.9	21.8	21.5	21.8	20.1	18.2	15.4	12.6	(20)	
2%		DB	14.3	16.2	20.3	24.1	28.1	33.1	34.5	34.7	29.9	24.4	18.5	14.9	(21)	
		MCWB	10.5	10.5	12.6	15.2	17.9	20.9	21.0	21.0	19.2	17.9	14.4	11.6	(22)	
5%		DB	12.9	14.5	18.6	22.0	26.2	31.2	33.0	32.7	27.9	22.9	17.1	13.5	(23)	
		MCWB	9.5	9.7	11.8	14.2	17.0	20.2	20.7	20.8	18.5	17.2	13.5	10.7	(24)	
10%		DB	11.4	12.9	16.8	20.1	24.4	29.3	31.6	31.0	26.1	21.3	15.7	12.1	(25)	
		MCWB	8.5	8.8	10.9	13.1	16.3	19.4	20.3	20.5	18.1	16.3	12.7	9.5	(26)	
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.3	12.2	14.4	17.3	20.6	23.6	23.4	23.6	21.7	19.8	16.5	13.3	(27)
MADB			15.4	16.6	20.1	24.7	28.3	33.7	31.8	31.1	27.5	23.8	19.0	15.5	(28)	
2%		WB	11.0	11.2	13.2	15.9	18.9	22.1	22.6	22.7	20.7	18.8	15.0	12.1	(29)	
		MADB	13.8	15.1	18.8	22.9	26.0	30.4	31.1	30.5	26.5	23.0	17.7	14.4	(30)	
5%		WB	10.0	10.3	12.4	14.7	17.9	21.1	21.8	22.0	19.8	17.9	14.0	11.1	(31)	
		MADB	12.2	13.7	17.4	20.6	24.3	29.1	30.2	29.7	25.5	21.7	16.5	13.0	(32)	
10%		WB	8.9	9.2	11.5	13.7	16.9	20.1	21.0	21.2	18.9	16.9	13.0	9.9	(33)	
		MADB	11.0	12.3	15.9	18.8	22.9	27.8	29.5	28.6	24.4	20.3	15.3	11.6	(34)	
(35)	Mean Daily Temperature Range	5% DB	MDBR	7.9	9.1	10.6	11.1	11.4	12.3	12.8	12.7	11.5	9.8	8.2	7.6	(35)
MCDBR			9.9	12.4	13.7	14.5	14.6	15.4	15.3	16.1	14.9	12.0	10.1	9.1	(36)	
MCWBR			6.7	7.6	7.4	7.2	6.5	6.1	5.4	5.6	6.3	6.4	6.6	6.2	(37)	
5% WB		MADB	9.0	11.3	12.4	13.1	13.5	14.6	14.4	14.6	13.0	10.8	8.9	8.1	(38)	
		MCWBR	6.5	7.3	7.1	6.9	6.6	6.2	5.7	5.8	6.2	6.2	6.2	5.9	(39)	
(40)	Clear-Sky Solar Irradiance	taub	0.321	0.342	0.381	0.398	0.407	0.406	0.399	0.397	0.385	0.387	0.348	0.318	(40)	
taud		2.454	2.392	2.297	2.276	2.300	2.329	2.349	2.362	2.383	2.368	2.430	2.472	(41)		
Ebn at Noon		800	844	850	865	865	865	867	857	838	778	756	766	(42)		
Edh at Noon		73	92	115	127	128	124	121	115	104	93	75	66	(43)		
(44)	All-Sky Solar Radiation	RadAvg	1.65	2.60	3.87	5.03	6.03	7.04	7.04	6.05	4.61	2.89	1.81	1.43	(44)	
(45)		RadStd	0.18	0.26	0.35	0.43	0.47	0.34	0.36	0.24	0.29	0.28	0.23	0.16	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	+0.94	N/A	N/A	N/A	N/A	N/A	N/A	(46)
(47)	Regional (36 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page